



to store the image, since there is a 4GB limit to the maximum size of a file on the FAT32 file-system. Hence you should use NTFS or Ext4 file systems that can handle such file size.

We'll be using a program called GNU ddrescue, this is primarily made for Linux, but is also available on Windows through cygwin. Go to bit.ly/tXPZy7 for instructions on how to install and use cygwin+ddrescue for Windows. It's present on most of the Linux distros, and is a part of System Rescue Disk that has been bundled in this month's Alpha DVD.

First identify the disk or partition that is giving problems using the command `sudo fdisk -l`. Say, if `/dev/sda` contains data to be recovered, you'll need to acquire another disk (or other media) onto which to save the output image. If it's only a partition of that particular drive, the path would look something like this `/dev/sda1`.

The following command illustrates how to image partition.

```
jalt@jalt:~$ sudo foremost -s 100 -t jpg -i /dev/sdb1
[sudo] password for jalt:
Processing: /dev/sdb1
|*****|
```

Foremost will not find any file if you've not selected the file types in the configuration file.

```
sudo ddrescue -r 3 /dev/sda1 <image> <log>
```

You can run successive passes like this:

```
sudo ddrescue -r 3 -C /dev/sda1 <image> <log>
```

GNU ddrescue will use the log file to only read the gaps with errors. In both cases, the `-r` option determines the number of times ddrescue will try to read when it encounters an error (`-1` = infinity). Another way to do this will be to copy as much data as possible, without retrying or splitting sectors:

```
sudo ddrescue --no-
```

```
jalt@jalt:~/recovery$ sudo scalpel /dev/sdb1 -o ./recovery
[sudo] password for jalt:
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/dev/sdb1"

Image file pass 1/2.
/dev/sdb1: 0.5% | 690.0 MB 2:25:09 ETA
```

Foremost's successor, Scalpel is said to give much better performance

```
split /dev/sda1 <image> <log>
```

Following this let it retry previous errors thrice, using uncached reads:

```
sudo ddrescue --direct --max-retries=3 /dev/hda1 <image> <log>
```

If that fails, you can try again but re-trimmed, so it tries to reread full sectors:

```
sudo ddrescue --direct --retrim --max-retries=3 /dev/hda1 <image> <log>
```

The second method will be redundant in case the device is not physically damaged. If you've just unknowingly deleted the files, then no need to run these commands, you can directly use the recovery programs. In case your media is corrupted and may contain

hardware defects, try making images by both the methods, run recovery on both the images. The data recovered may not be mutually exclusive, but will surely increase the number of files recovered. Now on to the actual recovery, you can use the programs that follow in any order you want or even all of them if you want.

Foremost

Foremost is a command-line tool which can recover files from a number of filesystems, including FAT, Ext4 and NTFS.

This tool works only in Linux. Foremost can recover files from an image of the drive, or from the drive directly. Consider the situation where an external hard disk at `/dev/sdb1` is damaged. To attempt an recovery from the hard drive, use the following commands:

```
mkdir ~/recovery
sudo foremost -i /dev/sdb1 -o ~/recovery/
```

To run Foremost on an image, just substitute the filename for the device

```
sudo foremost -i <image> -o ~/recovery/
```

The recovered files will then be owned by root. Change their ownership so that you can use them:

```
sudo chown -R youruser:youruser ~/recovery
```

Use the `-w` switch to obtain only an audit of recoverable files:

```
sudo foremost -w -i /dev/sdb1 -o ~/recovery
```

To recover only specific file types, use the `-t` switch:

```
sudo foremost -t jpg -i /dev/sdb1 -o ~/recovery
```

Scalpel

Scalpel is a fast file carver that reads a database of header and footer definitions and extracts matching files from a set of image files or raw device files. Based on Foremost, it's actually much more efficient. It's available for both Linux and Windows. For running it on Windows, you'll need "mingw" (MINimalist Gnu for Windows) installed. You can get scalpel from bit.ly/vnn9o7, once it is downloaded extract

it, head over to the src directory and use the command to install it.

```
mingw32-make -f Make-file.win
```

Once done, you'll also need to copy the pthreads and the regular expression library dlls into the same directory as "scalpel.exe". The default configuration file, "scalpel.conf", has all supported file patterns commented out, you must edit this file before running Scalpel to activate some patterns. Resist the urge to simply uncomment all file carving patterns; this wastes time and will generate a huge number of false positives. Instead, uncomment only the patterns for the file types you need. To run scalpel use the following command.

```
sudo scalpel FILE -o DIRECTORY
```

Where FILE is the image file (or device) and Directory is the output directory. Here the configuration file in the current directory is used. In case your configuration file resides somewhere else use the `-c` switch to specify the location of the configuration file.

PhotoRec

PhotoRec is a data recovery software designed to recover lost files including video, documents and archives from hard disks, CD-ROMs, and lost pictures (thus the Photo Recovery name) from digital camera memory. PhotoRec ignores the file system and goes after the underlying data, so it'll still work even if your media's file system has been severely damaged or formatted. PhotoRec is bundled in this month's Alpha DVD. There is no need for installation, extract the downloaded zip file and just run photorec_win.exe as an Administrator.

If you want to run PhotoRec on a disk image you'll have to specify which image